Introduction

Colour in a data visualisation is not decoration — it encodes information. The right palette makes ordered values read as ordered, signed values read as signed, and categorical values read as distinct without implying a ranking. The wrong palette (the infamous rainbow) creates false visual hotspots where the hue cycle bends and obscures structure that a perceptually uniform palette would reveal. Mastering the three palette families and choosing among them deliberately is one of the highest-payoff skills in figure design.

Prerequisites

A working knowledge of `ggplot2`'s colour and fill scales, and an awareness of colour-vision deficiencies that affect roughly 8 % of male readers.

Theory

Three palette families serve three different data types:

- **Sequential**: monotone hue or saturation gradient from low to high (`viridis`, `plasma`, `magma`, `YlOrRd`, `Blues`); for ordered data without a meaningful zero or centre.
- **Diverging**: two hues meeting at a neutral midpoint (`RdBu`, `PuOr`, `BrBG`); for signed data where positive and negative deviations from a centre are both meaningful.
- **Qualitative**: distinct, equally salient hues (`Set1`, `Set2`, `Dark2`, `Okabe-Ito`); for unordered categorical groups, ideally limited to eight or fewer.

`viridis` is the modern default for continuous scales because it is perceptually uniform (equal data steps look like equal visual steps), colour-blind safe under deuteranopia and protanopia, and degrades gracefully to greyscale. ColorBrewer palettes label colour-blind-friendly variants explicitly (`display.brewer.all(colorblindFriendly = TRUE)`), and the Okabe-Ito eight-colour palette was designed specifically for accessibility.

Assumptions

The data type matches the palette family — sequential for ordered values, diverging for signed, qualitative for nominal.

R Implementation

```r
library(ggplot2); library(viridis); library(RColorBrewer)

# Continuous: viridis
ggplot(faithfuld, aes(waiting, eruptions, fill = density)) +
  geom_tile() +
  scale_fill_viridis_c(option = "plasma") +
  theme_minimal()

# Continuous: diverging
ggplot(mtcars, aes(wt, mpg, colour = hp - mean(hp))) +
  geom_point(size = 3) +
  scale_colour_gradient2(low = "blue", mid = "grey80", high = "red",
                         midpoint = 0) +
  theme_minimal()

# Qualitative: Brewer Set2
ggplot(mtcars, aes(wt, mpg, colour = factor(cyl))) +
  geom_point(size = 3) +
```

```r
  scale_colour_brewer(palette = "Set2") +
  theme_minimal()

# Show colour-blind-safe palettes
display.brewer.all(colorblindFriendly = TRUE)
```

Output & Results

Four illustrative figures: a 2D density tile plot using `plasma` for sequential encoding, a centred-at-zero scatter using `gradient2` for signed deviations, a categorical scatter using Brewer `Set2` for cylinder counts, and a panel of all colour-blind-friendly Brewer palettes for reference.

Interpretation

A reporting sentence (figure caption): "Cell colour encodes 2D kernel density on the `viridis::plasma` perceptually uniform palette; positive and negative horsepower deviations from the mean use a red-blue diverging scale centred at zero." Always state the palette by name when it is non-default; readers should not have to guess.

Practical Tips

- Use `viridis` (or its variants `magma`, `plasma`, `inferno`, `cividis`) as the default for continuous scales; all are perceptually uniform and colour-blind safe.
- Limit qualitative palettes to 8 colours or fewer; beyond that, switch to shapes, linetypes, or facets to disambiguate groups.
- Use `scale_colour_gradient2()` (or `scale_fill_gradient2()`) for any signed quantity; the centred palette communicates symmetry that one-sided sequentials cannot.
- For greyscale fallback (photocopying, monochrome printing), `viridis` translates almost losslessly; rainbow hues collapse into similar greys and become unreadable.
- Verify colour-blind accessibility on every figure with `colorBlindness::cvdPlot()`; simulating deuteranopia and protanopia takes seconds and prevents costly resubmissions.
- For heatmaps, perceptual uniformity is non-negotiable — a non-uniform palette creates apparent hotspots wherever the hue cycle bends.

R Packages Used

`viridis` for `viridis`, `magma`, `plasma`, `inferno`, `cividis` continuous scales; `RColorBrewer` for the Brewer family of qualitative, sequential, and diverging palettes; `colorspace` for HCL-based custom palettes; `paletteer` for unified access to dozens of curated palettes from across the R ecosystem.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- ggplot2 grammar and multi-panel layouts